Project Part-1

Project Proposal

Project Partners

Please communicate with your partners often. Share contact information / establish the best way to communicate about the project. It is an expectation that you and your partner will contribute equally across all project components. Note: Just because you are in the same group doesn't mean you are entitled to the same grade. You will turn in a **peer evaluation form** in addition to the final project, which will factor into your final grade.

No late work will be accepted. If you have a documented emergency that prevents you from completing a the project, please contact your instructor and provide proof of the emergency.

Purpose

The main purpose of the project proposal is to help you think about the project early, so you can get a head start on finding data, practice reading in data, thinking about the questions you wish to answer, etc.

The goal of the final project will be to ask meaningful questions on a data set you are interested in, and use the computing tools learned in class to help explore your questions.

Instructions

Identify 1 data set you're interested in using for the final project. If you're unsure where to find data, you can use the list of potential data sources in the Resources for data sets section as a starting point. It may also help to think of topics you're interested in investigating and find data sets on those topics.

Criteria for datasets

The data sets should meet the following criteria:

- At least 100 observations (or approved by your instructor)
- At least 5 unique columns that are actual variables (or approved)

  Identifier variables such as "name", "social security number", etc. are not useful explanatory variables.

  If you have multiple columns with the same information (e.g. "state abbreviation" and "state name"), then they are not unique explanatory variables.
- Data must be real (if you are unsure, do not choose these data)
- Data sets must have a mix of quantitative and categorical variables

Resources for data

You can find data wherever you like, but here are some recommendations to get you started. You shouldn't feel constrained to data sets that are already in a tidy format, you can start with data that needs cleaning and tidying.

1. The UCI machine learning repository: https://archive.ics.uci.edu/
2. Kaggle: https://www.kaggle.com

   Note: There are many fake data sets on Kaggle. Do not choose these.
3. data.gov
4. data.europa.eu

If you have other data in mind that are not from these sources, please make sure that these data are real, and you are interested in analyzing them. Note: The data can not be from a R package/ already built into SAS. You need to be able to download the data and read the data in appropriately.

What you will turn in

Your project proposal will consist of the following sections with the following information.

Introduction and data section

For your data set:

- Identify the source of the data.
- State when and how it was originally collected (by the original data curator, not how you found the data). If this information is not available, state that and write 1-2 sentences on how that may impact your findings.
- Write a brief description of the observations.
- Explain why you are interested in these data.

Research Question

You are tasked to write ONE research question.

Before writing your research questions, please see the following handout

Your research questions should contain AT LEAST 3 variables. For example: What is the relationship between a penguin's bill length and their flipper length for each of the species in our data set?

Your research question should be clear and easily understood by a general audience. Again, when writing a research question, please think about the following:

- What is your target audience?
- Can the question be answered?

This section in your proposal should contain:

- A well formulated research question
- Statement on why this question is important / matters
- Identification if the variables you are working with are categorical or quantitative

Display your data

For your data set:

- Read the data in correctly
- Use the `glimpse()` function to provide a glimpse of the data set

An example of a properly glimpsed data set can be seen below:

```
Rows: 344
Columns: 8
$ species <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
$ island <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
$ bill_length_mm <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
$ bill_depth_mm <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
$ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
$ body_mass_g <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
$ sex <fct> male, female, female, NA, female, male, female, male~
$ year <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

Notice that I can clearly see that these data have 8 columns, 344 rows, and have no obvious abnormalities with the data itself.

How to turn in

You should turn in a well formatted PDF generated from Quarto with the following subsections to Gradescope.

(Level - 1 header): Introduction and data

(Level - 2 header): Research question

(Level - 4 header): Why this matters and identification of variable types

Proposal Grading

Component	Points
Total	40 pts
Introduction and data	15
Research Question	15
Why this matters and identification of variable types	10